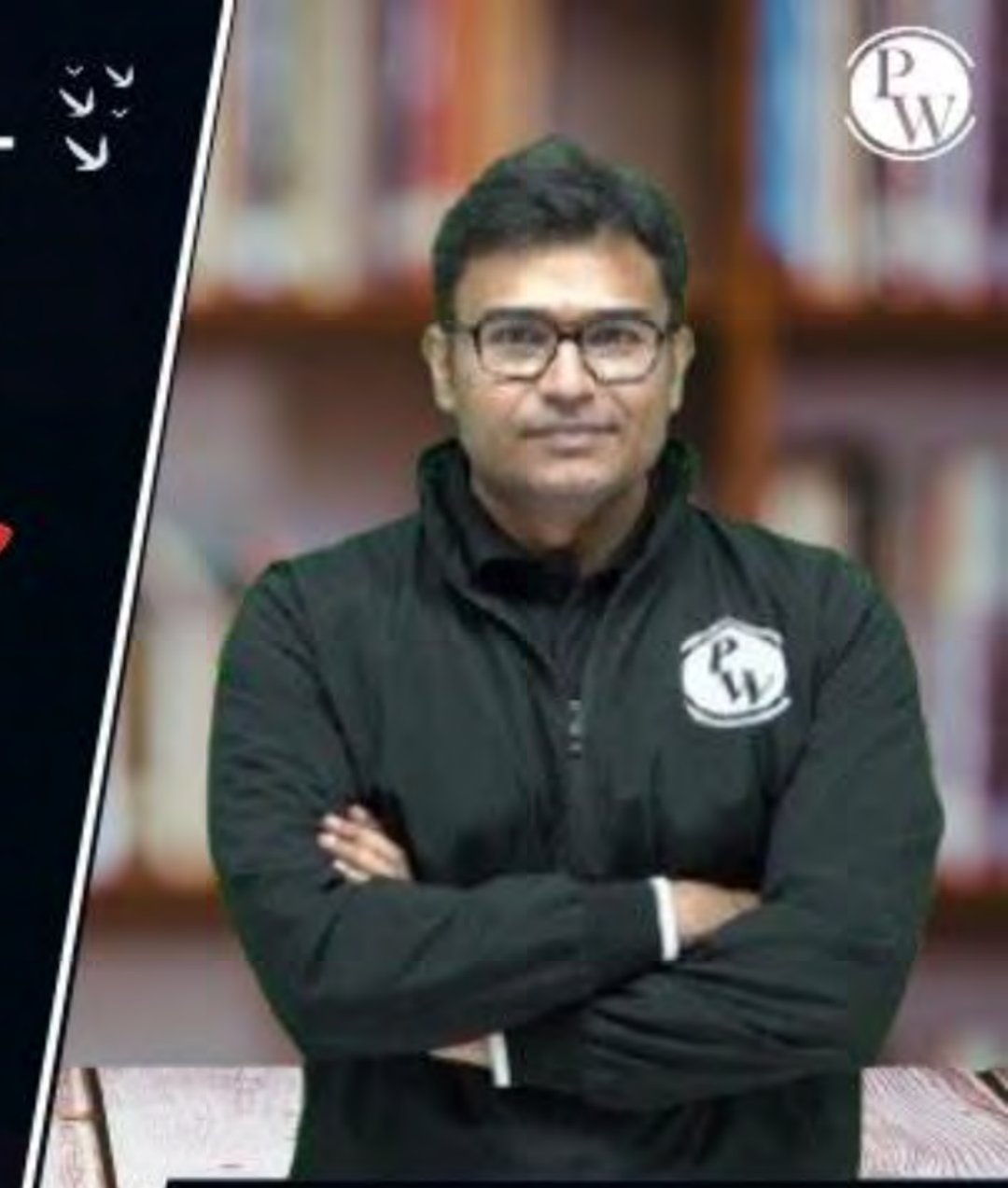
Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 2



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

DFA, NFA, RE

Topics to be Covered



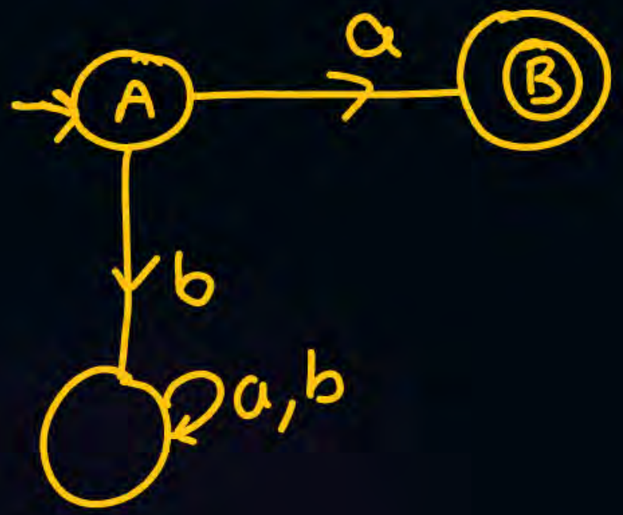
Topic

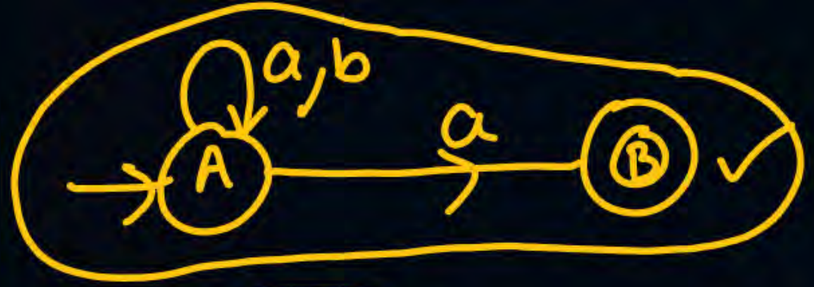
Topic

Topic

NFA, DFA

DFA



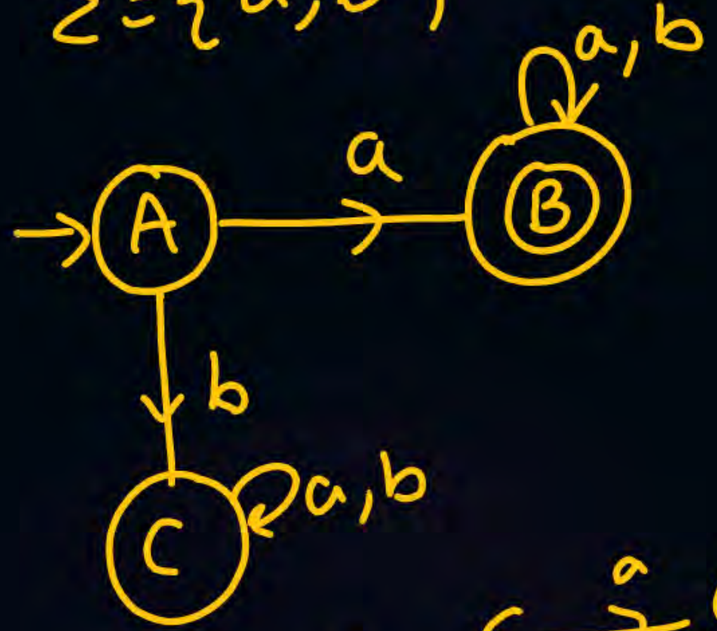




Every dfa is an nfa

There are some NFAS which are not DFA

$\Sigma = \{a, b\}$



$A \xrightarrow{a} B$

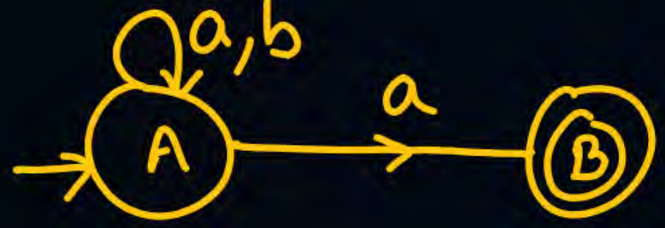
$A \xrightarrow{b} C$

$B \xrightarrow{a} B$

$B \xrightarrow{b} B$

$C \xrightarrow{a} C$

$C \xrightarrow{b} C$



$A \xrightarrow{a} A, B$ ✓

$A \xrightarrow{b} A$

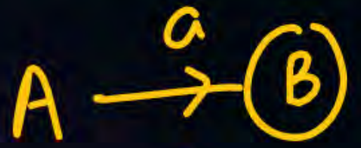
$B \xrightarrow{a} \phi$

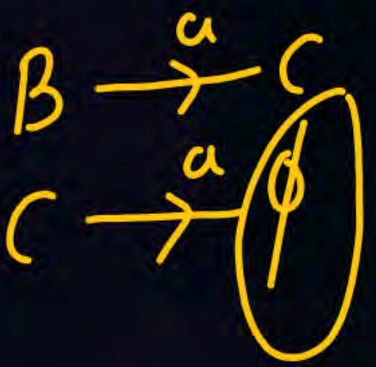
$B \xrightarrow{b} \phi$ ✓

$\Sigma = \{a\}$



✓







✗ DFA



Both DFA & NFA are equal in power

If there is an NFA for a language, then there is sure a DFA for the langage.

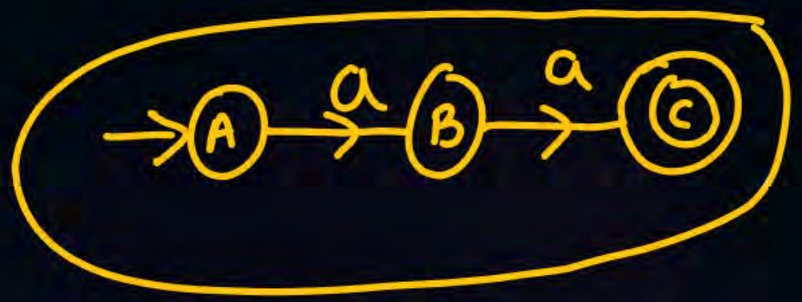
There is an algorithm to convert NFA into DFA.


Language
NFA
DFA
RE
N1
N2
N3
minimal NFA not unique
D1
D2
D3
minimal DFA unique
R1
R2
R3
minimal X


Language
DFA, NFA, RE

$\Sigma = \{a\}$ ✓ $\Sigma^* = \{\epsilon, a, aa, aaa \ldots\ldots\}$

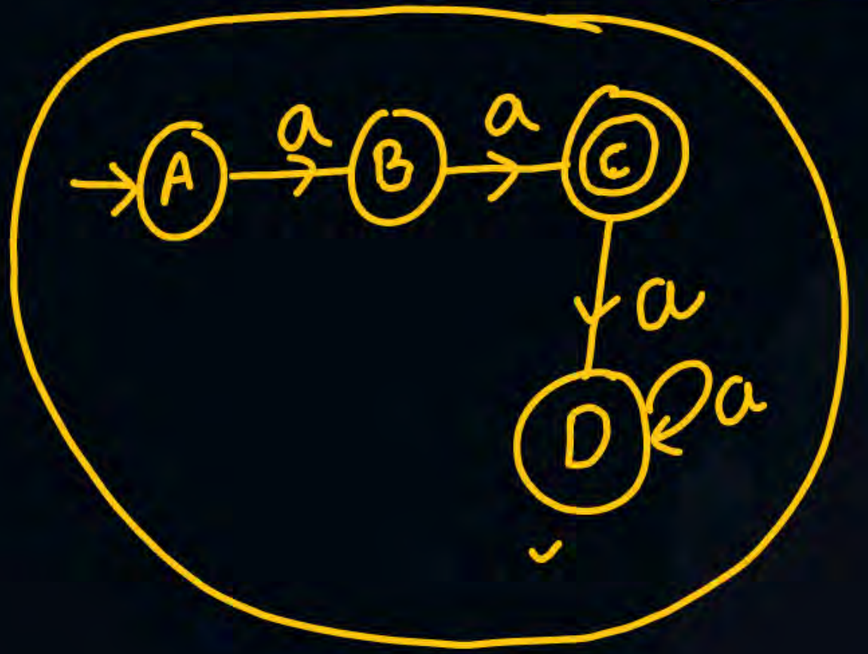
$L = \{aa\}$ $RE = aa$

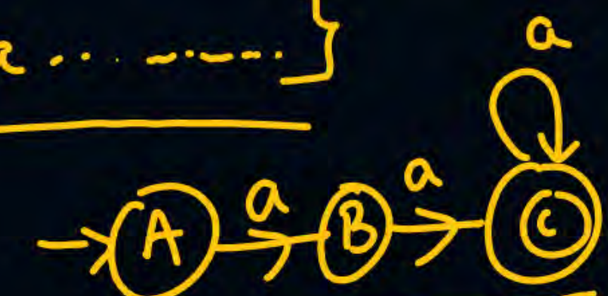


min NFA

$C \xrightarrow{a} \phi$

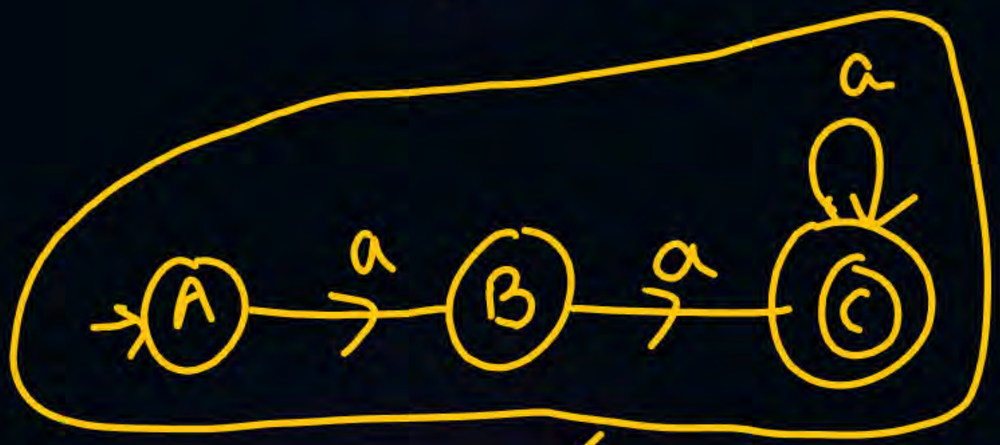
min DFA



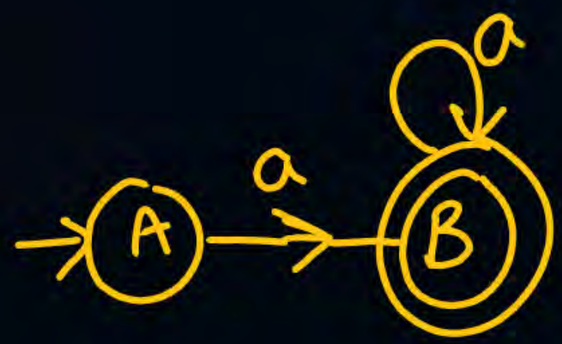


2 ✓
3
4
.....

$L = \{aa\}$



X



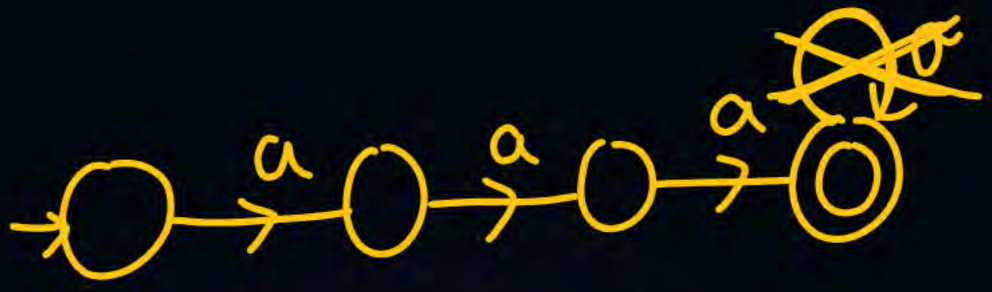
$A \xrightarrow{a} B \xrightarrow{a} C$

$A \xrightarrow{a} B \xrightarrow{a} C \xrightarrow{a} C$

2
3
4
5
⋮

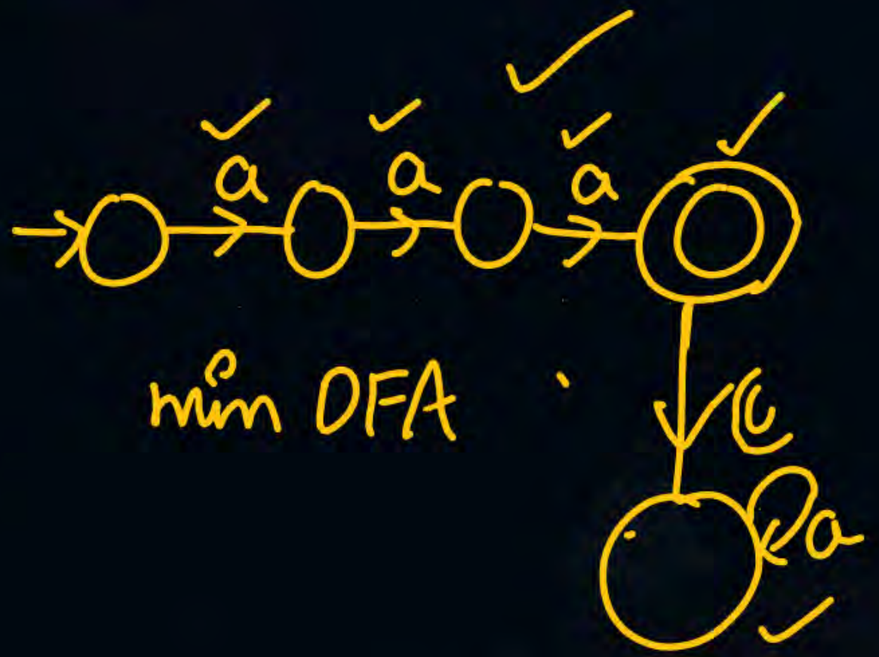
1
2
3
4
⋮

$\Sigma = \{a\}$ $L = \{aaa\}$ RE = aaa

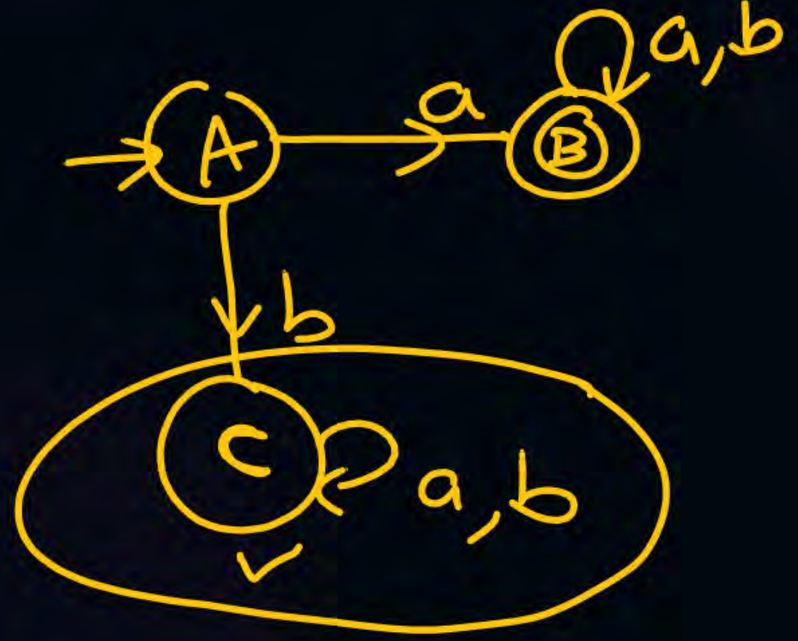


min NFA





min DFA


A
a
B
a,b
b
C
a,b

$\Sigma = \{a\}$ $L = \{aa, aaa\}$ RE = aa + aaa



min NFA

T



DFA & NFA:

more than one final state possible

Initial state can be final state

Dead state always ① in min DFA

min NFA has 0 Dead states

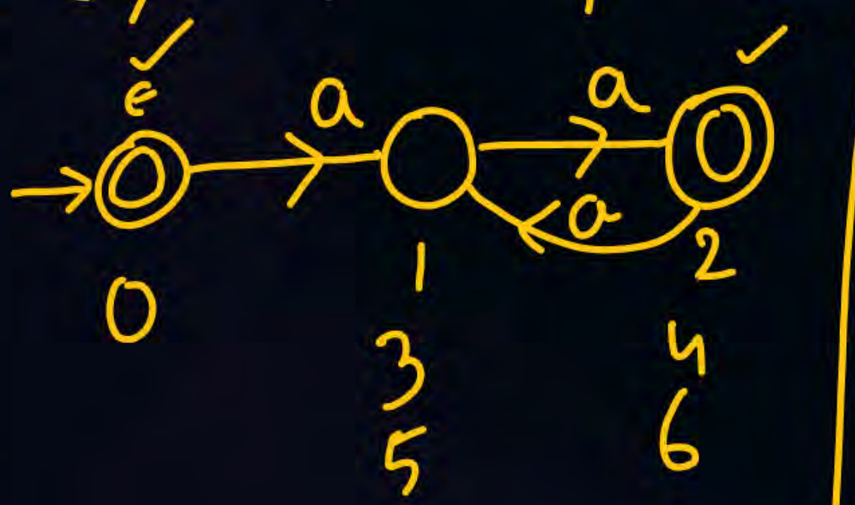
$\Sigma = \{a\}$ $L = \{\omega / n_a(\omega) \text{ are even}\}$

ϵ, aa, aaaa, aaaaaa RE: $(aa)^*$

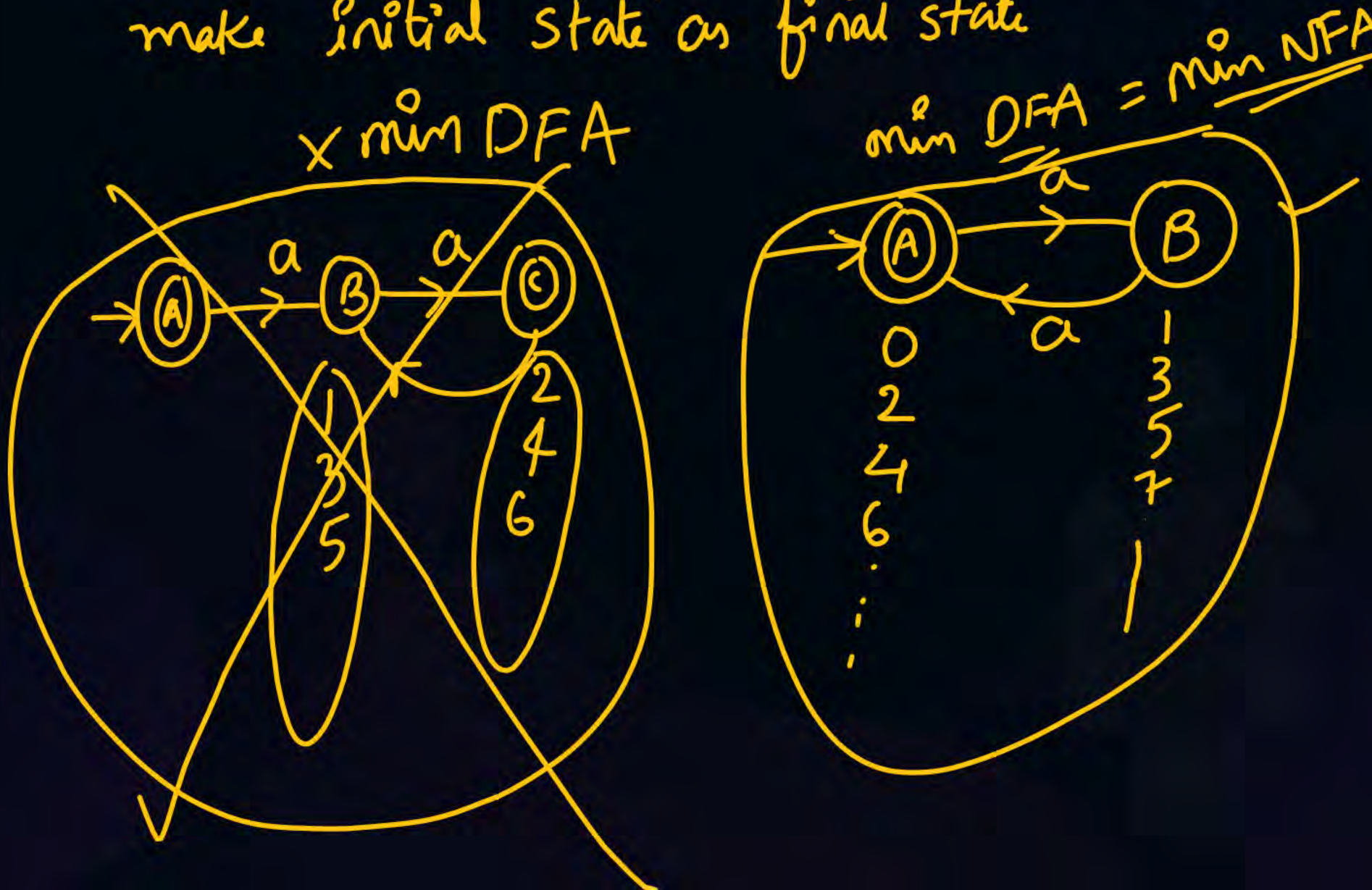
$(aa)^* = (aa)^0, (aa)^1, (aa)^2, (aa)^3 \ldots\ldots\ldots$

$= \epsilon, aa, aaaa, aaaaaa, \ldots\ldots$

ϵ → Initial

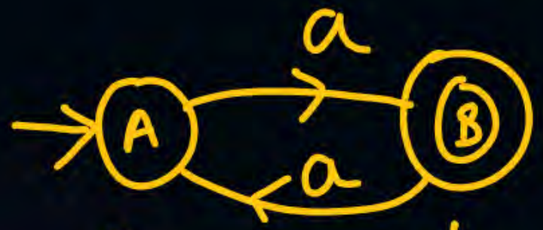


If ϵ has to accepted by NFA & DFA, make initial state as final state



$\Sigma = \{a\}$ Odd no of as $= \{a, aaa, aaaaa, \cdots\}$

$RE = (aa)^*a = a(aa)^*$



min DFA
= min NFA

0
2
4
6

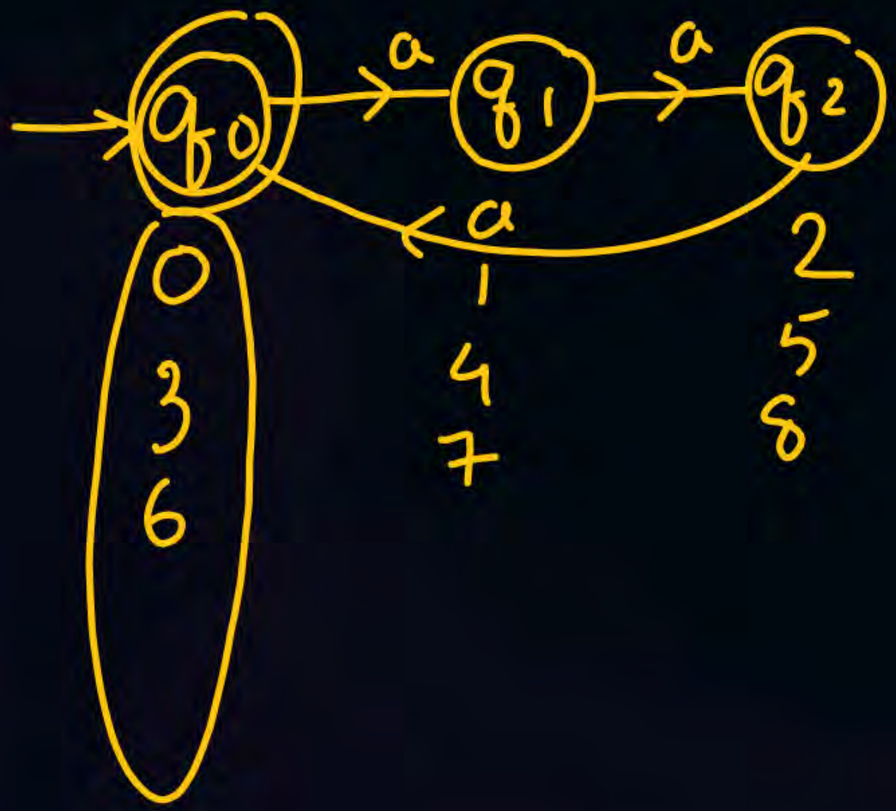
1
3
5
7

$\Sigma = \{a\}$ $L = \{ w / n_a(w) \text{ is divisible by } 3 \}$



$RE = (aaa)^*$

min DFA = min NFA

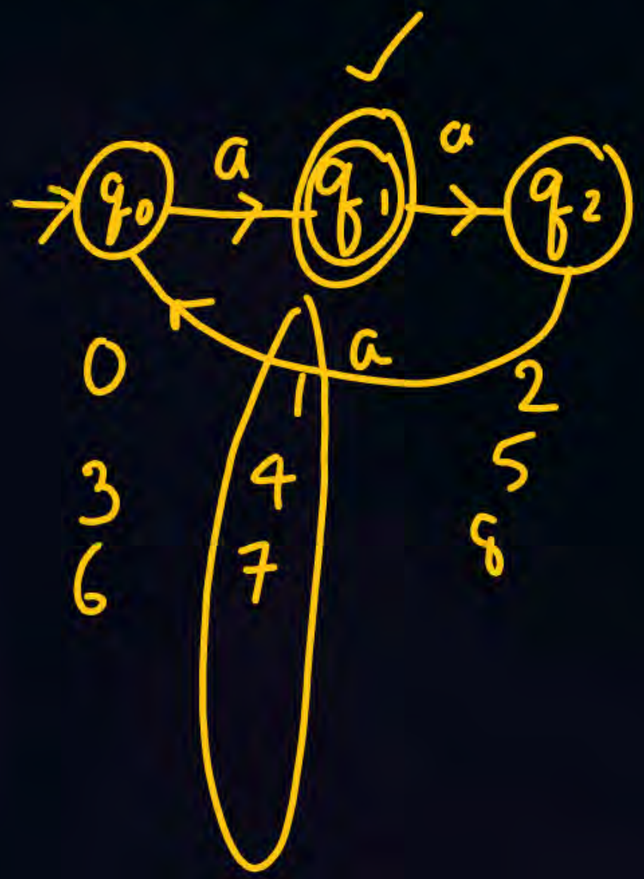


$\Sigma = \{a\}$ $L = \{w / n_a(w) \equiv (1) \bmod 3\}$

$1, 4, 7, 10 \ldots\ldots$



$RE = (aaa)^*a = a(aaa)^*$



$\Sigma = \{a\}$ $L = \{w / n_a(w) \cong 2 \bmod 3\}$

$RE = \underline{aa(aaa)^*} = (aaa)^*aa = a(aaa)^*a$



min DFA
= min NFA

$aa(aaa)^*$

a^{3n+2} ✓

0	1	2
3	4	5
6	7	8

$\Sigma = \{a\}$ $L = \{a^n / n = 3 \text{ to } 5\} = \{aaa, aaaaa\}$

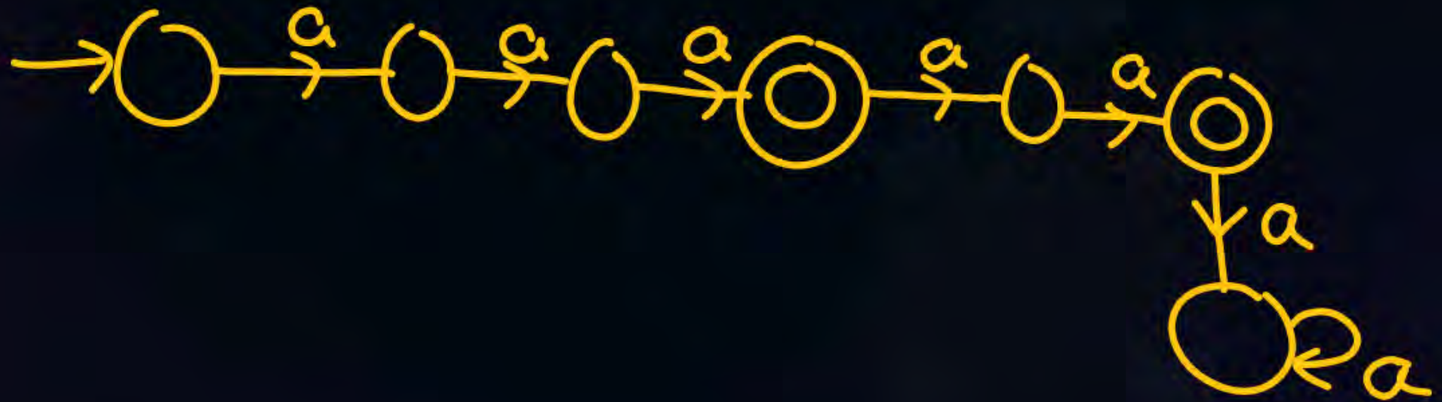
RE = aaa + aaaaa

min NFA



min NFA

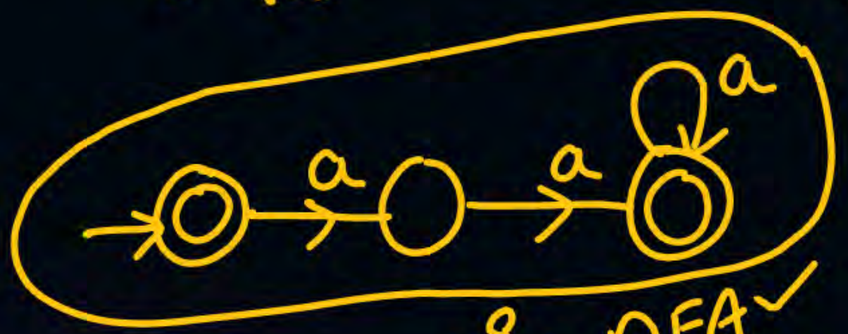
min DFA



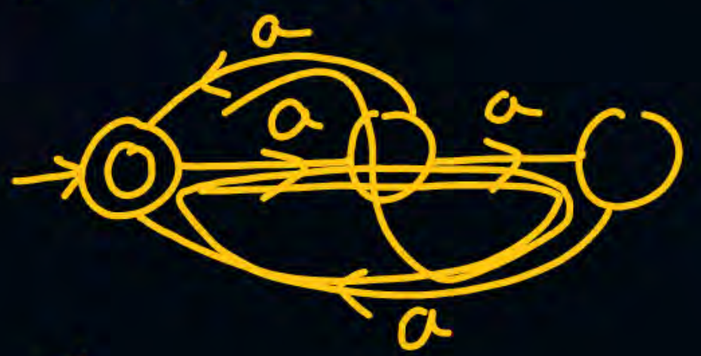
$L = (aa + aaa)^* = \{\epsilon, a^2, a^3, a^4, a^5, a^6, a^7 \text{ ---------- }\}$

$2 \quad 2 \quad 2 \quad 3 \quad 3 \quad 3 \quad 2 \quad 2 \quad 3$

RE



min DFA = min NFA



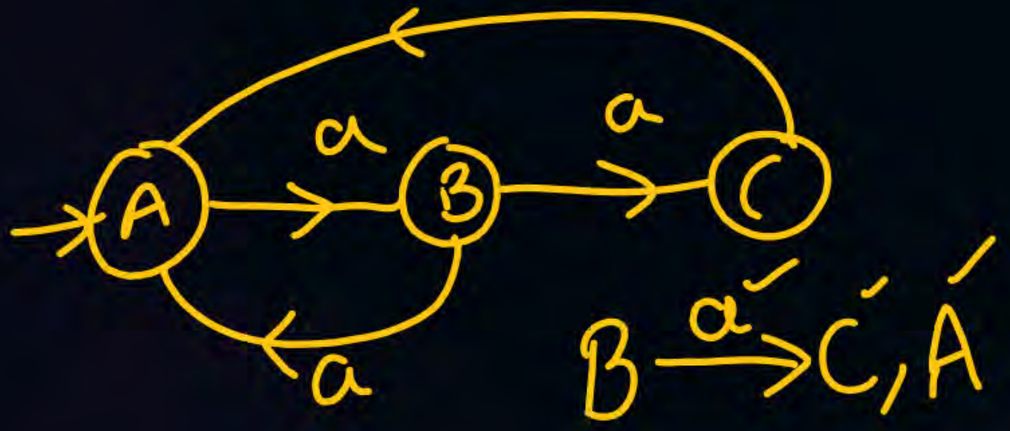
$L = \{(aa + aaa)^*$

$= a^n / n = 2x + 3y, x, y \geq 0$

a^5

$aaaaa$


a
a
a
a


a
a
A
B
C
a
B $\xrightarrow{a}$ C, A

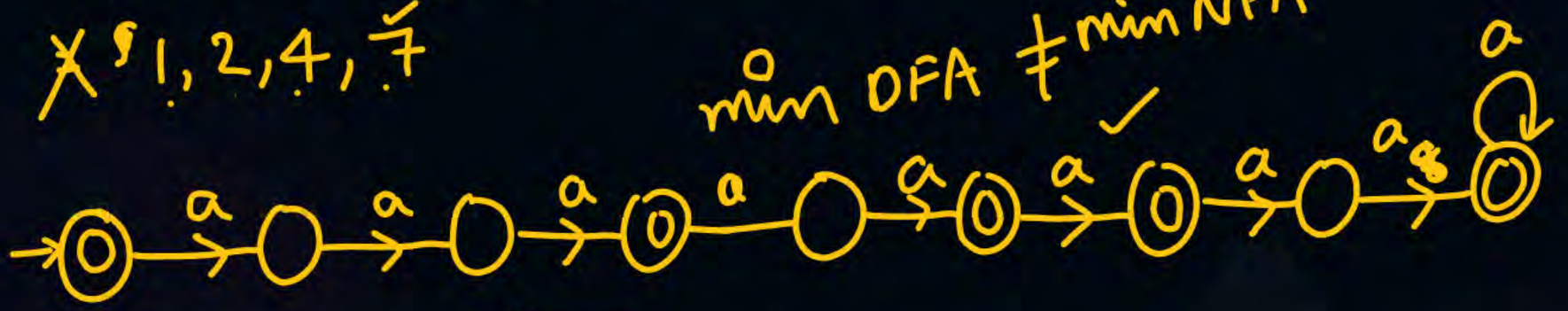
$L = (aaa + aaaaa)^*$

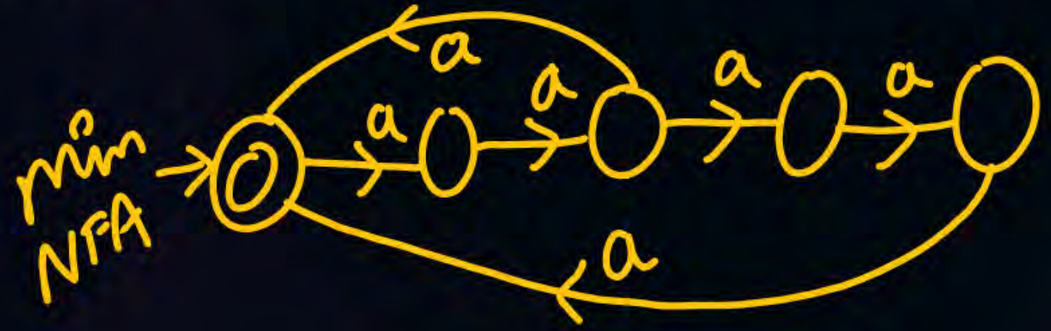
$L = \{\epsilon, 3, 5, 6, 8, 9, 10, 11, \ldots\ldots\}$

$3 = 3+3$, $5 = 3+5$, $6 = 3+3$, $8 = 3+5$, $9 = 3+3+3$, $10 = 5+5$, $11 = 3+3+5$

$\times$ 1, 2, 4, 7

min DFA $\neq$ min NFA





$$|min(NFA)| <= |min(DFA)|$$

$$(aaaa + aaaaa)^*$$

THANK - YOU